Principal Component Analysis

Principal Component Analysis

- Principal Component Analysis is an unsupervised learning algorithm used for the dimensionality reduction in machine learning
- In Principal Component Analysis (PCA), the goal is to transform a large set of variables into a smaller set that still contains most of the information.
- It is a projection method where data with m-columns (features) is projected into a subspace with m or fewer columns, whilst retaining the essence of the original data.
- It is a statistical process that **converts the observations of correlated features into a set of linearly uncorrelated features with the help of orthogonal transformation.** These new transformed features are called the **Principal Components**.
- The number of principal components is <= the number of original variables or the number of observations
- This transformation is defined in such a way that the first principal component has the largest possible variance (ie, accounts for as much of the variability in the data as possible) and each succeeding component in turn has the highest variance possible under the constraint that it is orthogonal to the preceding components

Why need PCA?

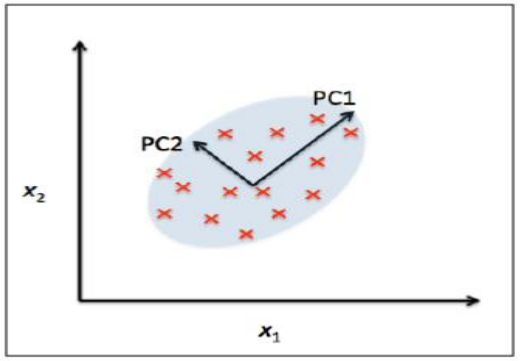
- **Principal Component Analysis (PCA)** is used mainly to **simplify data while retaining as much important information as possible**.

1. **1. Dimensionality Reduction**
    1. Real-world datasets often have many features (high dimensions).
    2. PCA reduces them to fewer principal components while preserving maximum variance.
    3. This makes models faster and easier to train.
2. **2. Remove Redundancy (Correlation)**
    1. Many features are highly correlated.
    2. PCA transforms them into uncorrelated (orthogonal) components.
    3. This avoids duplication of information.
3. **3. Noise Reduction**
    1. Small-variance components often represent noise.
    2. PCA can discard these, keeping only important patterns.
4. **5. Data Visualization**
    1. PCA helps reduce data to lower dimensions, making it easy to visualize patterns, clusters, and outliers.
5. **6. Handles Curse of Dimensionality**
    1. In high dimensions, data becomes sparse.
    2. PCA reduces dimensions → improves distance-based methods like KNN, clustering.

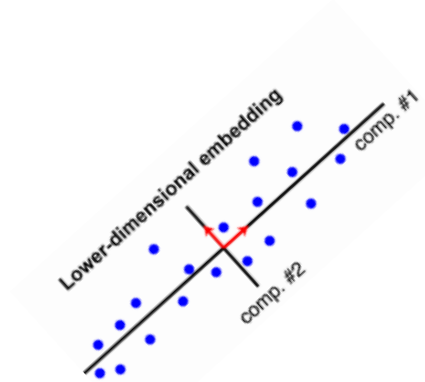
Key Characteristics

- **Uncorrelated:** Unlike the original variables, which might be highly correlated, the new PCs are completely uncorrelated with each other.
- **Information Density:** PC1 always accounts for the largest possible variance in the data, followed by PC2, and so on.
- **Dimensionality Reduction:** Because the later PCs often capture very little variance (mostly noise), you can often "drop" them, effectively reducing the dimensions of your dataset while keeping the most important patterns.

What PCA does?



- PCA works by identifying the directions (principal components) where there is the most "spread" or **variance** in your data.
- A principal component is a synthetic, orthogonal axis (or linear combination of original variables) created to capture the maximum possible variance in a high-dimensional dataset.
- When you project high-dimensional data onto principal components, you reduce dimensionality while retaining as much informative variance as possible.
- Imagine you have a 2D scatter plot of data points shaped like an oval. PCA tries to find the straight line that best describes the data's shape.
  - **Principal Component 1 (PC1):** This is the line that goes through the center of the data and captures the maximum amount of variation. If you "squashed" all your data points onto this line, they would be spread out as much as possible.
  - **Principal Component 2 (PC2):** This line is **orthogonal** (at a 90-degree angle) to PC1 and captures the remaining variance.



- Thus, PCA identifies a list of the principal axes to describe the underlying dataset before ranking them according to the amount of variance captured by each.
- https://www.youtube.com/watch?v=g-Hb26agBFg

Principal components?

Geometrically, each Principal Component (PC) is a **line** (or hyperline) in feature space onto which the data points are orthogonally projected.

Each principal component is a linear combination of the observed variables (features), where the coefficients are given by the **eigenvectors of the covariance matrix**.

$$PC1 = w_{1,1}(\text{Feature A}) + w_{2,1}(\text{Feature B}) + w_{3,1}(\text{Feature C}).... + w_{n,1}(\text{Feature N})$$

$$PC2 = w_{1,2}(\text{Feature A}) + w_{2,2}(\text{Feature B}) + w_{3,2}(\text{Feature C}).... + w_{n,2}(\text{Feature N})$$

$$PC3 = w_{1,3}(\text{Feature A}) + w_{2,3}(\text{Feature B}) + w_{3,3}(\text{Feature C}).... + w_{n,3}(\text{Feature N})$$

Here, Ws are the eigen vectors, computed by solving an **eigenvalue problem on the covariance matrix**.

- **Eigenvectors (w):** Determine the **direction** (the orientation of the new axes).
- **Eigenvalues (lambda):** Determine the **magnitude** (how much variance or spread is actually captured along each of those directional axes).

- **Principal components=Eigen vectors with largest eigen values**
- Number of PCs= the number of features in the dataset. For example, a dataset with ten features can have max. ten eigenvalues/eigenvector combinations.

Core Mathematical & Geometric Properties

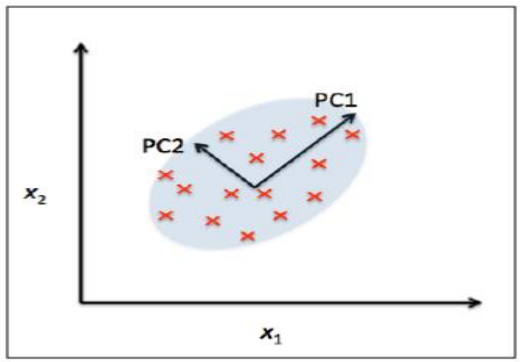
1. **Linear Combination of Features**
   Each principal component (PC_i) is calculated as a weighted linear combination of the original centered/standardized features ($X_1, X_2, \ldots, X_n$):

$$PC_i = w_{i1}X_1 + w_{i2}X_2 + \cdots + w_{in}X_n$$

   The weights (w_{ij}) represent the **eigenvectors** of the data's covariance (or correlation) matrix, often referred to as **loadings**.

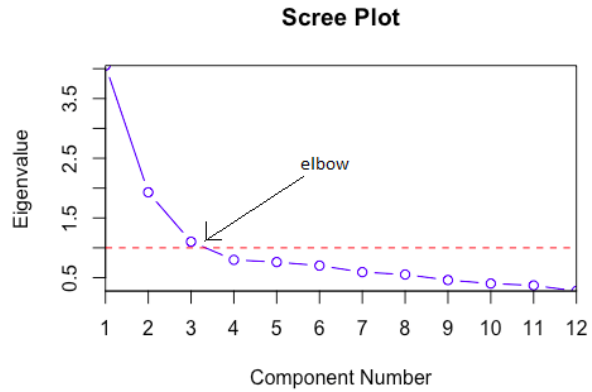
2. **Orthogonality (Uncorrelated Axes)**
   By construction, every subsequent principal component is mathematically orthogonal (perpendicular) to all preceding components. This guarantees that the derived components are completely **uncorrelated** with one another, eliminating multicollinearity.

3. **Maximum Variance Ranking**
   - PC_1 **(First Principal Component):** Axis aligned along the direction of the *maximum possible variance* in the dataset.
   - PC_2 **(Second Principal Component):** Axis orthogonal to PC_1 that captures the *second highest amount of remaining variance*, and so on.

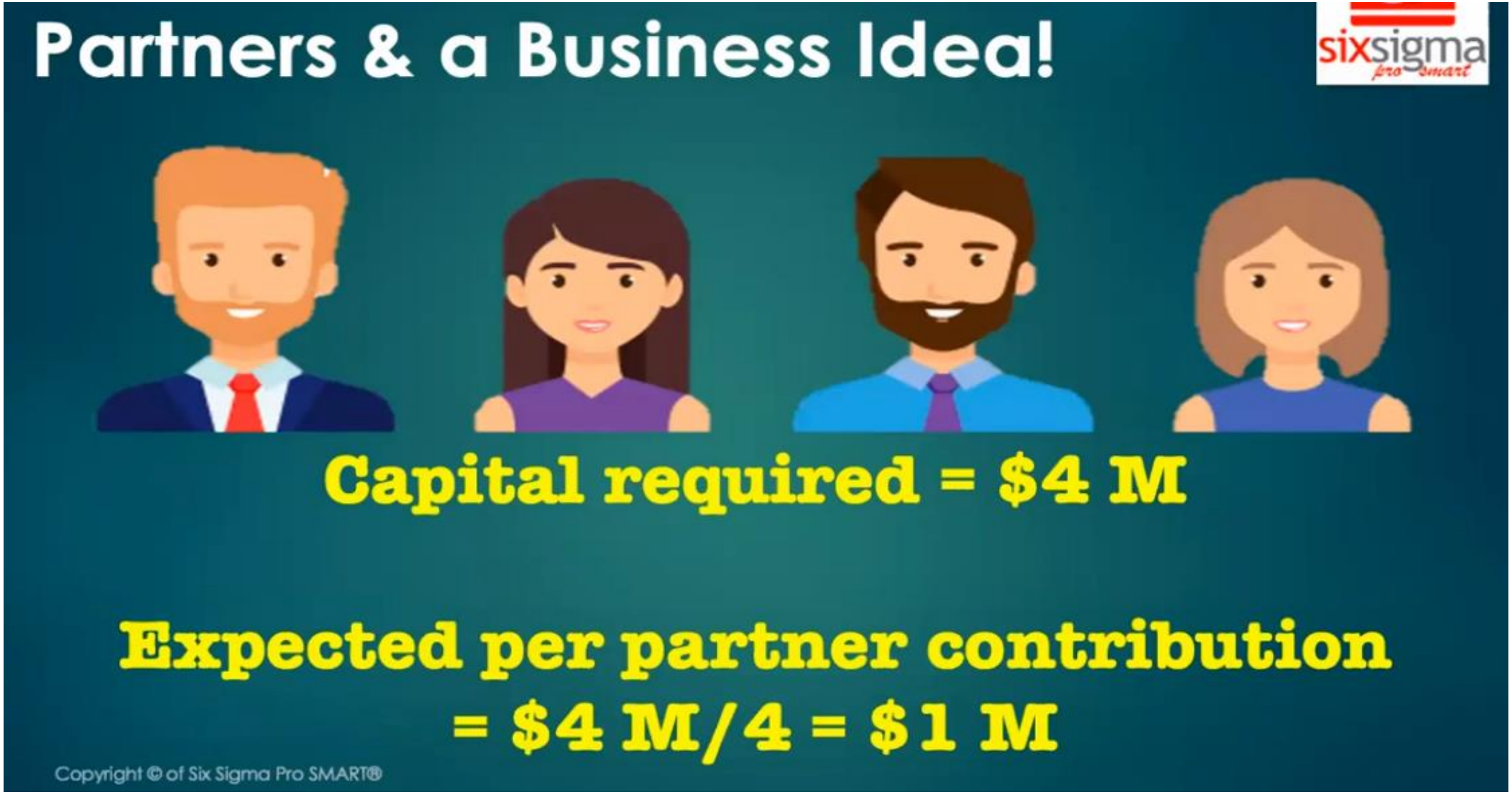


Geometrically, each Principal Component (PC) is a **line** (or hyperline) in feature space onto which the data points are orthogonally projected.

- How Many PCs to Keep?
  - We **don't keep all PCs** — only the most important ones. Such as:
    - **First PC** → maximum variance direction
    - **Second PC** → next highest variance (orthogonal to first)
    - And so on...
  - Common methods:
    - (a) **Explained Variance Method:**
      - Each PC explains some % of total variance.
      - Choose enough PCs to cover **~90%–95% variance**.
      - Example:
        - PC1 → 60%
        - PC2 → 25%
        - PC3 → 10%
        - Total = 95% → Keep **first 3 PCs**
    - **(b) Scree Plot Method**
      - Plot: PC index vs eigenvalues
      - Choose PCs before the "elbow point"
    - **(c) Eigenvalue > 1 Rule (Kaiser Criterion)**
      - Keep PCs whose eigenvalue > 1



Example:
https://www.youtube.com/watch?v=wWt6cAHlDbg



Partners & a Business Idea!

Expected per partner contribution
= $4 M/4 = $1 M



Actual contribution	1.8	1.2	0.6	0.4
	✓	✓	X	X

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Partners & a Business Idea!






	1.8	1.2	0.6	0.4
Proportion %	45%	30%	15%	10%
Cumulative %	45%	75%	90%	100%

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Eigen Values

Eigen values represent the units of spread captured by each PC.

PCs are extracted in the descending order of Eigen Values.

Who is more important?



	1.8	1.2	0.6	0.4
Proportion %	45%	30%	15%	10%
Cumulative %	45%	75%	90%	100%

Interpreting PCA output

Observed variables		
LDL	Hypertension	Glucose
169	185	143
154	194	247
185	179	319
183	192	129
205	210	390
192	210	394
151	173	369
150	150	393
199	196	270
145	141	353
	-------More data------	

PC1	PC2
0.9LDL + 0.75Hypertension -0.25Glucose	0.1LDL + 0.06Hypertension +0.93Glucose

$$PC1 = a_1x_1 + a_2x_2 + a_3x_3$$
$$PC2 = b_1x_2 + b_2x_2 + b_3x_3$$
$$PC3 = c_1x_1 + c_2x_2 + c_3x_3$$

PC1 = 0.9X169 + 0.75X185 - 0.25X143 = 255.10

PC2 = 0.1X169 + 0.06X185 + 0.93X143 = 160.99

Reduced dimensions

Observed variables		
LDL	Hypertension	Glucose
169	185	143
154	194	247
185	179	319
183	192	129
205	210	390
192	210	394
151	173	369
150	150	393
199	196	270
145	141	353
	------More data------	

PC1	PC2
255.1	160.99
222.35	256.75
221	325.91
276.45	149.79
244.5	395.8
231.8	398.22
173.4	368.65
149.25	389.49
258.6	282.76
148	351.25
---More data---	

Steps Involved:

Step 1. Data

We consider a dataset having n features or variables denoted by $X_1, X_2, \ldots, X_n$. Let there be N examples. Let the values of the i-th feature X_i be $X_{i1}, X_{i2}, \ldots, X_{iN}$ (see Table 4.1).

Features	Example 1	Example 2	$\cdots$	Example N
X_1	X_{11}	X_{12}	$\cdots$	X_{1N}
X_2	X_{21}	X_{22}	$\cdots$	X_{2N}
$\vdots$				
X_i	X_{i1}	X_{i2}	$\cdots$	X_{iN}
$\vdots$				
X_n	X_{n1}	X_{n2}	$\cdots$	X_{nN}

Table 4.1: Data for PCA algorithm

- PCA is used to identify the components with the maximum variance, and the contribution of each variable to a component is based on its magnitude of variance. It is best practice to normalize the data before conducting a PCA as unscaled data with different measurement units can distort the relative comparison of variance across features.

Step 2. Compute the means of the variables

We compute the mean $\bar{X}_i$ of the variable X_i:

$$\bar{X}_i = \frac{1}{N}(X_{i1} + X_{i2} + \cdots + X_{iN}).$$

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Step 3. Calculate the covariance matrix

Consider the variables X_i and X_j (i and j need not be different). The covariance of the ordered pair (X_i, X_j) is defined as[1]

$$\mathrm{Cov}\,(X_i, X_j) = \frac{1}{N-1}\sum_{k=1}^{N}(X_{ik} - \bar{X}_i)(X_{jk} - \bar{X}_j). \tag{4.1}$$

We calculate the following $n \times n$ matrix S called the covariance matrix of the data. The element in the i-th row j-th column is the covariance $\mathrm{Cov}\,(X_i, X_j)$:

$$S = \begin{bmatrix} \mathrm{Cov}\,(X_1, X_1) & \mathrm{Cov}\,(X_1, X_2) & \cdots & \mathrm{Cov}\,(X_1, X_n) \\ \mathrm{Cov}\,(X_2, X_1) & \mathrm{Cov}\,(X_2, X_2) & \cdots & \mathrm{Cov}\,(X_2, X_n) \\ \vdots & & & \\ \mathrm{Cov}\,(X_n, X_1) & \mathrm{Cov}\,(X_n, X_2) & \cdots & \mathrm{Cov}\,(X_n, X_n) \end{bmatrix}$$

The aim of this step is to understand how the variables of the input data set are varying from the mean with respect to each other, or in other words, to see if there is any relationship between them.
The covariance matrix is a $p \times p$ symmetric matrix (where p is the number of dimensions) that has as entries the covariances associated with all possible pairs of the initial variables.

What do the covariances that we have as entries of the matrix tell us about the correlations between the variables?

It is the sign of the covariance that matters :

- if positive then : the two variables increase or decrease together (correlated)
- if negative then : One increases when the other decreases (Inversely correlated)

Step 4. Calculate the eigenvalues and eigenvectors of the covariance matrix

Let S be the covariance matrix and let I be the identity matrix having the same dimension as the dimension of S.

i) Set up the equation:

$$\det(S - \lambda I) = 0. \tag{4.2}$$

This is a polynomial equation of degree n in λ. It has n real roots (some of the roots may be repeated) and these roots are the eigenvalues of S. We find the n roots $\lambda_1, \lambda_2, \ldots, \lambda_n$ of Eq. (4.2).

The story behind eigen values and eigen vectors:

https://www.youtube.com/watch?v=5UjQVJu89_Q

PCA 5: finding eigenvalues and eigenvectors - YouTube

ii) If $\lambda = \lambda'$ is an eigenvalue, then the corresponding eigenvector is a vector

$$U = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

such that

$$(S - \lambda' I)U = 0.$$

(This is a system of n homogeneous linear equations in $u_1, u_2, \ldots, u_n$ and it always has a nontrivial solution.) We next find a set of n orthogonal eigenvectors $U_1, U_2, \ldots, U_n$ such that U_i is an eigenvector corresponding to λ_i.[2]

iii) We now normalise the eigenvectors. Given any vector X we normalise it by dividing X by its length. The length (or, the norm) of the vector

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

is defined as

$$||X|| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Given any eigenvector U, the corresponding normalised eigenvector is computed as

$$\frac{1}{||U||}U.$$

We compute the n normalised eigenvectors $e_1, e_2, \ldots, e_n$ by

$$e_i = \frac{1}{||U_i||}U_i, \quad i = 1, 2, \ldots, n.$$

Eigen vectors represent the direction in which PCs are aligned

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Step 5. Derive new data set

Order the eigenvalues from highest to lowest. The unit eigenvector corresponding to the largest eigenvalue is the first principal component. The unit eigenvector corresponding to the next highest eigenvalue is the second principal component, and so on.

i) Let the eigenvalues in descending order be $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$ and let the corresponding unit eigenvectors be $e_1, e_2, \ldots, e_n$.

ii) Choose a positive integer p such that $1 \leq p \leq n$.

iii) Choose the eigenvectors corresponding to the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_p$ and form the following $p \times n$ matrix (we write the eigenvectors as row vectors):

$$F = \begin{bmatrix} e_1^T \\ e_2^T \\ \vdots \\ e_p^T \end{bmatrix},$$

where T in the superscript denotes the transpose.

PCA 6: coordinates in low-dimensional space - YouTube

iv) We form the following $n \times N$ matrix:

$$X = \begin{bmatrix} X_{11} - \bar{X}_1 & X_{12} - \bar{X}_1 & \cdots & X_{1N} - \bar{X}_1 \\ X_{21} - \bar{X}_2 & X_{22} - \bar{X}_2 & \cdots & X_{2N} - \bar{X}_2 \\ \vdots & & & \\ X_{n1} - \bar{X}_n & X_{n2} - \bar{X}_n & \cdots & X_{nN} - \bar{X}_n \end{bmatrix}$$

v) Next compute the matrix:

$$X_{\text{new}} = FX.$$

Note that this is a $p \times N$ matrix. This gives us a dataset of N samples having p features.

Step 6. New dataset

The matrix X_{new} is the new dataset. Each row of this matrix represents the values of a feature. Since there are only p rows, the new dataset has only features.

Problem

Given the data in Table 4.2, use PCA to reduce the dimension from 2 to 1.

Feature	Example 1	Example 2	Example 3	Example 4
X_1	4	8	13	7
X_2	11	4	5	14

Table 4.2: Data for illustrating PCA

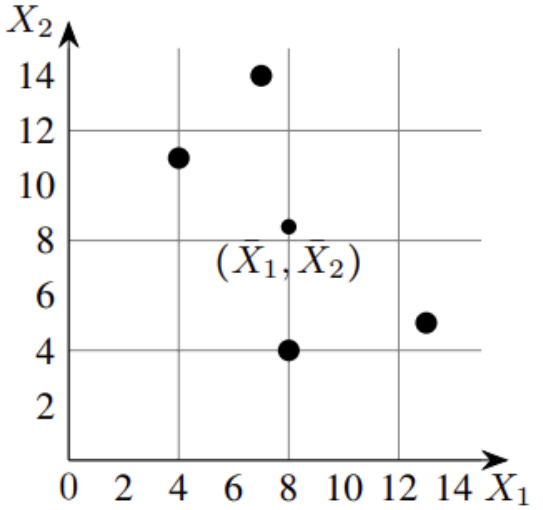


Figure 4.2: Scatter plot of data in Table 4.2

Solution

1. Compute Mean

We have

$$\bar{X}_1 = \frac{1}{4}(4 + 8 + 13 + 7) = 8,$$
$$\bar{X}_2 = \frac{1}{4}(11 + 4 + 5 + 14) = 8.5.$$

Figure 4.2 shows the scatter plot of the data together with the point $(\bar{X}_1, \bar{X}_2)$.

PCA problem / How to compute principal components / KTU Machine learning

2. Calculation of the covariance matrix

The covariances are calculated as follows:

$$\begin{aligned}
\mathrm{Cov}(X_1, X_1) &= \frac{1}{N-1}\sum_{k=1}^{N}(X_{1k} - \bar{X}_1)^2 \\
&= \tfrac{1}{3}\left((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2\right) \\
&= 14 \\
\mathrm{Cov}(X_1, X_2) &= \frac{1}{N-1}\sum_{k=1}^{N}(X_{1k} - \bar{X}_1)(X_{2k} - \bar{X}_2) \\
&= \tfrac{1}{3}((4-8)(11-8.5) + (8-8)(4-8.5) \\
&\quad + (13-8)(5-8.5) + (7-8)(14-8.5) \\
&= -11 \\
\mathrm{Cov}(X_2, X_1) &= \mathrm{Cov}(X_1, X_2) \\
&= -11 \\
\mathrm{Cov}(X_2, X_2) &= \frac{1}{N-1}\sum_{k=1}^{N}(X_{2k} - \bar{X}_2)^2 \\
&= \tfrac{1}{3}\left((11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2\right) \\
&= 23
\end{aligned}$$

The covariance matrix is

$$S = \begin{bmatrix} \mathrm{Cov}(X_1, X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Cov}(X_2, X_2) \end{bmatrix} = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

ance matrix captures how
es are related

3. Eigenvalues of the covariance matrix

The characteristic equation of the covariance matrix is

$$
\begin{aligned}
0 &= \det(S - \lambda I) \\
&= \begin{vmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{vmatrix} \\
&= (14 - \lambda)(23 - \lambda) - (-11) \times (-11) \\
&= \lambda^2 - 37\lambda + 201
\end{aligned}
$$

Solving the characteristic equation we get

$$
\begin{aligned}
\lambda &= \tfrac{1}{2}(37 \pm \sqrt{565}) \\
&= 30.3849,\ 6.6151 \\
&= \lambda_1,\ \lambda_2 \quad \text{(say)}
\end{aligned}
$$

$Av = \lambda v$

where,

- *A is the matrix,*
- *v is associated eigenvector and*
- *\lambda is scalar eigenvalue.*

4. Computation of the eigenvectors

To find the first principal components, we need only compute the eigenvector corresponding to the largest eigenvalue. In the present example, the largest eigenvalue is λ_1 and so we compute the eigenvector corresponding to λ_1.

The eigenvector corresponding to $\lambda = \lambda_1$ is a vector $U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ satisfying the following equation:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = (S - \lambda_1 I).U$$

$$= \begin{bmatrix} 14 - \lambda_1 & -11 \\ -11 & 23 - \lambda_1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$= \begin{bmatrix} (14 - \lambda_1)u_1 - 11u_2 \\ -11u_1 + (23 - \lambda_1)u_2 \end{bmatrix}$$

This is equivalent to the following two equations:

$$(14 - \lambda_1)u_1 - 11u_2 = 0$$
$$-11u_1 + (23 - \lambda_1)u_2 = 0$$

Using the theory of systems of linear equations, we note that these equations are not independent and solutions are given by

$$\frac{u_1}{11} = \frac{u_2}{14 - \lambda_1} = t,$$

that is

$$u_1 = 11t, \quad u_2 = (14 - \lambda_1)t,$$

where t is any real number. Taking $t = 1$, we get an eigenvector corresponding to λ_1 as

$$U_1 = \begin{bmatrix} 11 \\ 14 - \lambda_1 \end{bmatrix}.$$

To find a unit eigenvector, we compute the length of X_1 which is given by

$$\begin{aligned}\|U_1\| &= \sqrt{11^2 + (14 - \lambda_1)^2} \\ &= \sqrt{11^2 + (14 - 30.3849)^2} \\ &= 19.7348\end{aligned}$$

Therefore, a unit eigenvector corresponding to $lambda_1$ is

$$\begin{aligned}e_1 &= \begin{bmatrix} 11/\|U_1\| \\ (14 - \lambda_1)/\|U_1\| \end{bmatrix} \\ &= \begin{bmatrix} 11/19.7348 \\ (14 - 30.3849)/19.7348 \end{bmatrix} \\ &= \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}\end{aligned}$$

By carrying out similar computations, the unit eigenvector e_2 corresponding to the eigenvalue $\lambda = \lambda_2$ can be shown to be

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}.$$

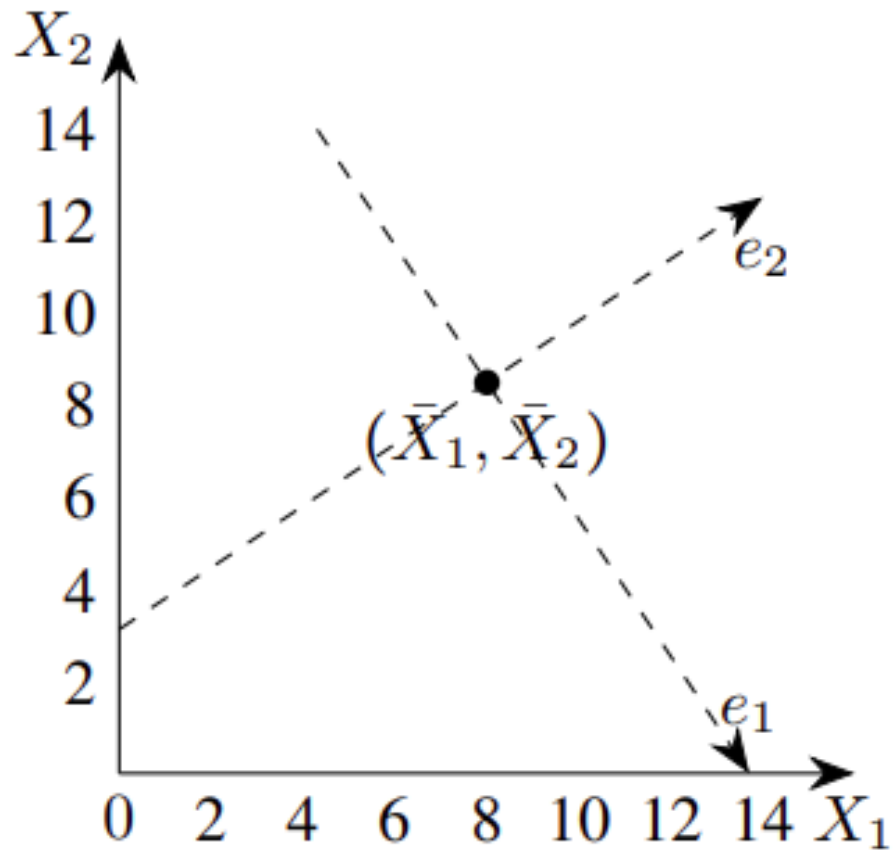


Figure 4.3: Coordinate system for principal components

5. Computation of first principal components

Let $\begin{bmatrix} X_{1k} \\ X_{2k} \end{bmatrix}$ be the k-th sample in Table 4.2. The first principal component of this example is given by (here "T" denotes the transpose of the matrix)

$$e_1^T \begin{bmatrix} X_{1k} - \bar{X}_1 \\ X_{2k} - \bar{X}_2 \end{bmatrix} = \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} X_{1k} - \bar{X}_1 \\ X_{2k} - \bar{X}_2 \end{bmatrix}$$
$$= 0.5574(X_{1k} - \bar{X}_1) - 0.8303(X_{2k} - \bar{X}_2).$$

For example, the first principal component corresponding to the first example $\begin{bmatrix} X_{11} \\ X_{21} \end{bmatrix} = \begin{bmatrix} 4 \\ 11 \end{bmatrix}$ is calculated as follows:

$$\begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} X_{11} - \bar{X}_1 \\ X_{21} - \bar{X}_2 \end{bmatrix} = 0.5574(X_{11} - \bar{X}_1) - 0.8303(X_{21} - \bar{X}_2)$$
$$= 0.5574(4 - 8) - 0.8303(11 - 8,5)$$
$$= -4.30535$$

The results of calculations are summarised in Table 4.3.

X_1	4	8	13	7
X_2	11	4	5	14
First principal components	-4.3052	3.7361	5.6928	-5.1238

Table 4.3: First principal components for data in Table 4.2

6. Geometrical meaning of first principal components

As we have seen in Figure 4.1, we introduce new coordinate axes. First we shift the origin to the "center" $(\bar{X}_1, \bar{X}_2)$ and then change the directions of coordinate axes to the directions of the eigenvectors e_1 and e_2 (see Figure 4.3).

Next, we drop perpendiculars from the given data points to the e_1-axis (see Figure 4.4). The first principal components are the e_1-coordinates of the feet of perpendiculars, that is, the projections on the e_1-axis. The projections of the data points on e_1-axis may be taken as approximations of the given data points hence we may replace the given data set with these points. Now, each of these approximations can be unambiguously specified by a single number, namely, the e_1-coordinate of approximation. Thus the two-dimensional data set given in Table 4.2 can be represented approximately by the following one-dimensional data set (see Figure 4.5):

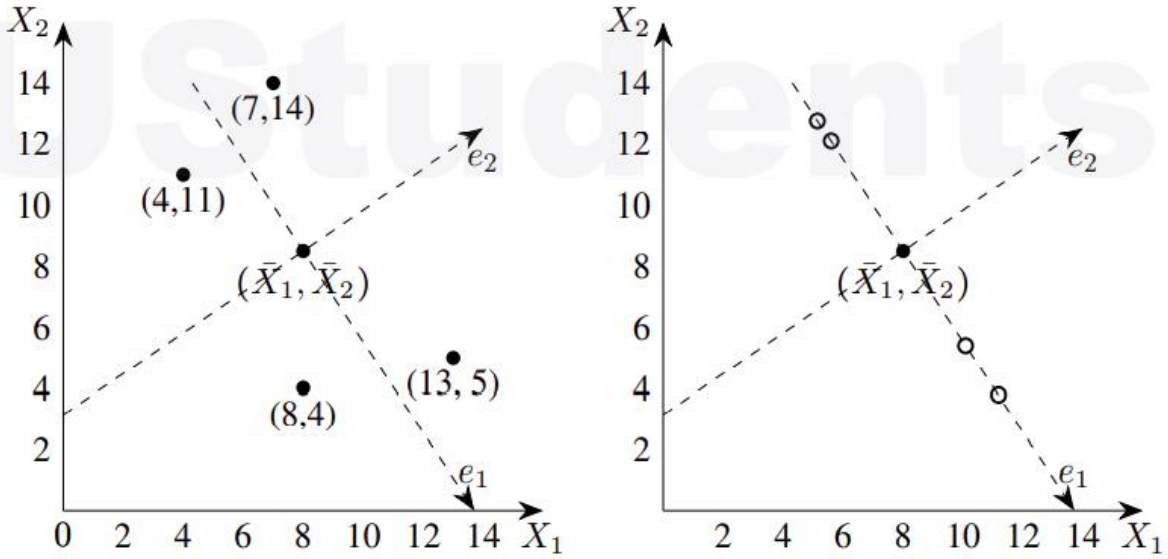


Figure 4.5: Geometrical representation of one-dimensional approximation to the data in Table 4.2

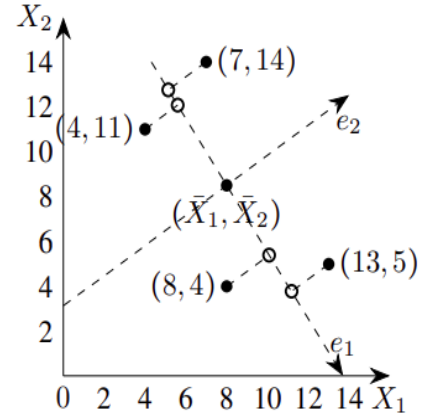


Figure 4.4: Projections of data points on the axis of the first principal component

PC1 components	-4.305187	3.736129	5.692828	-5.123769

Table 4.4: One-dimensional approximation to the data in Table 4.2

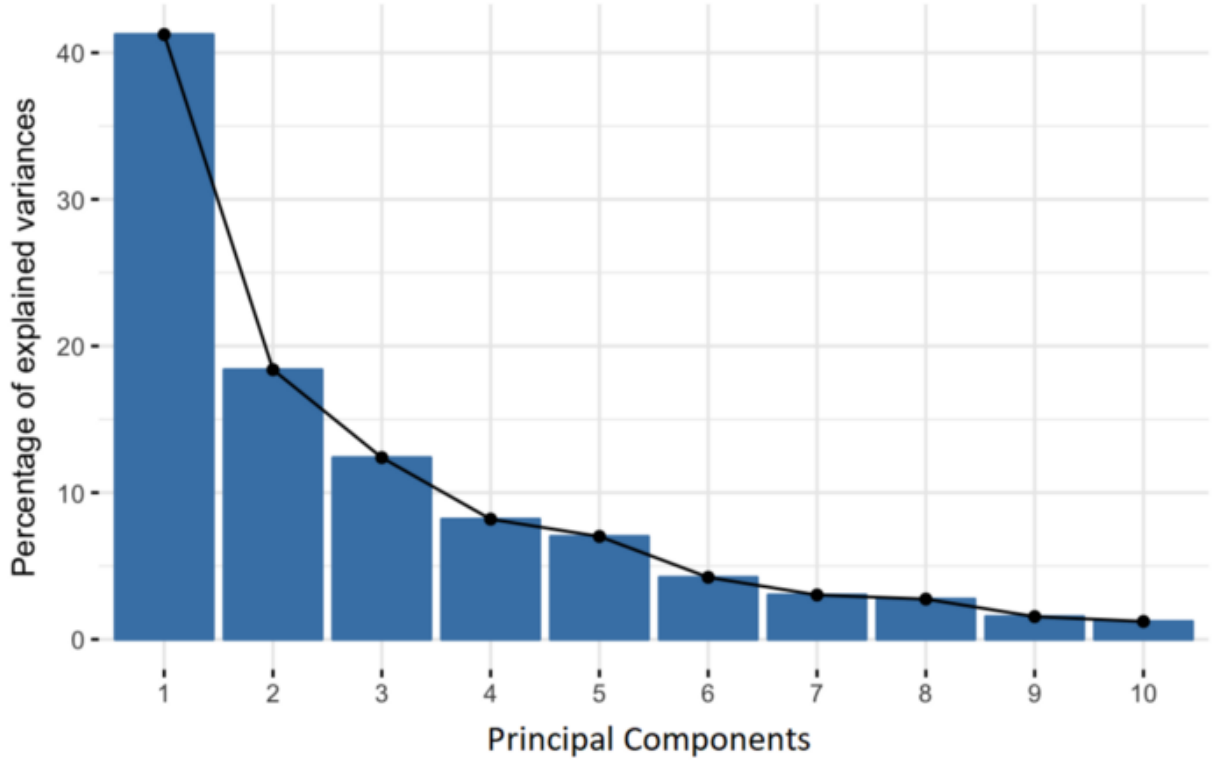
Each step in the computation answers:

Step	Question it answers
Centering	Where is the true shape of data?
Covariance	How are variables related?
Eigenvectors	What are the best directions?
Eigenvalues	Which directions matter most?
Projection	How do we represent data efficiently?

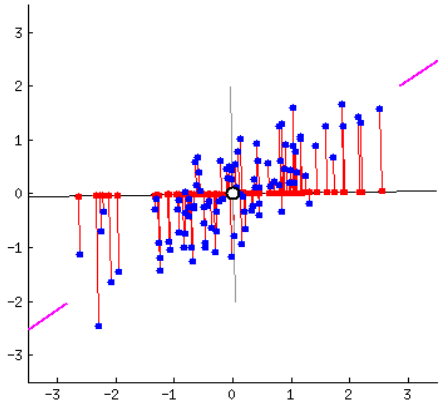
Consider the following dataset containing the values of two features X1 and X2 for four observations. Apply Principal Component Analysis (PCA) to reduce the dimensionality of the dataset from two dimensions to one dimension.

Observation	X1	X2
A	2	0
B	0	2
C	3	1
D	1	3

- principal components are, the lines that capture most information of the data.
- principal components are constructed in such a manner that the first principal component accounts for the **largest possible variance** in the data set.



- can we guess the first principal component ?
- Yes, it's approximately the line that matches the purple marks because it goes through the origin and it's the line in which the projection of the points (red dots) is the most spread out.
- The second principal component is calculated in the same way, with the condition that it is uncorrelated with (i.e., perpendicular to) the first principal component and that it accounts for the next highest variance.



- PCA 3: direction of greatest variance – YouTube
- PCA 4: principal components = eigenvectors - YouTube

The story behind eigen values and eigen vectors:
https://www.youtube.com/watch?v=5UjQVJu89_Q
PCA 5: finding eigenvalues and eigenvectors - YouTube